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★ 29,000+ GitHub Stars

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RESEARCH SUMMARY

When something goes wrong, someone has to be able to say what happened and who is answerable. AI systems that act on their own can make that hard: the damage shows up, the path to it often does not. My lab builds three things: tools that catch unsafe permissions before an agent ships, a checkpoint that inspects tool calls before they run, and records good enough to piece together afterward who did what. This work spans **auditable AI systems** across four audit questions: *auditing of action* (auditability and control of agent systems across their lifecycle), *optimization of effort* (agent efficiency, routing, and the cost of defense), *verification of output* (trust and robustness evaluation of foundation models), and *detection in data* (anomaly and out-of-distribution detection). Auditing of action, verification of output, and detection in data ask whether something departed from what was expected and whether anyone can still tell. Optimization of effort asks whether the work was necessary at all. Beyond a system's behavior, my lab studies its real-world impact: how LLMs and agents can affect people and the decisions made with them. The current frontier is **agent auditability**: deployable open-source infrastructure that audits agents from before deployment to after a run.

1. Auditing of Action: Auditability and Control of Agent Systems

When a system acts through tools, APIs, and shared state, what it does becomes the primary failure surface. This thread covers auditability frameworks, runtime control surfaces, agent-specific failure modes, and post-run attribution and recovery, spanning the full lifecycle from pre-deployment to after a run.

- ☐ Agent Auditability
- ☐ Agent Systems
- ☐ Multi-Agent Systems
- ☐ Behavior Verification
- ☐ AI Agent Security
- ☐ Runtime Guardrails
- ☐ Policy Enforcement

2. Optimization of Effort: Agent Efficiency, Routing, and the Cost of Defense

The record that makes a system accountable also shows what its effort bought. This thread asks whether the work was necessary: what defense training costs an agent's ability to finish a task, how model choice becomes an online cost-quality decision, and how large-scale detection cuts its compute cost. It is the dimension industry asks about most.

- ☐ Agent Efficiency
- ☐ LLM Routing
- ☐ Cost-Quality Trade-offs
- ☐ Workflow Generation
- ☐ Cost of Defense Training
- ☐ Scalable Detection

3. Verification of Output: Trust and Robustness of Foundation Models

Between the input and the action, foundation models add their own failure surface: behavior under attack, hallucination, and adversarial robustness of retrieval. Failures on these surfaces can propagate downstream when model output is passed to an agent or a user-facing system. My lab measures these surfaces through jailbreak detection for vision-language models, causal analysis of hallucination, retrieval-augmented generation attacks, and LLM-as-anomaly-detector benchmarks.

- ☐ Foundation-Model Trust
- ☐ Trustworthiness Benchmarks
- ☐ LLM Hallucination
- ☐ Jailbreak Detection
- ☐ RAG Security
- ☐ Adversarial Robustness
- ☐ Multimodal Trust

4. Detection in Data: Anomaly and Out-of-Distribution Detection

Here the question is whether an input departed from the conditions represented during development and evaluation. This is my methodological and open-source base, built through the PyOD ecosystem and ADBench. It continues through benchmarks for NLP, automatic OOD detector selection, modality-specific OOD methods for graphs, multimodal data, and video, and few-shot cross-domain OOD detection. When an incident involves distribution shift, these methods supply evidence about whether unfamiliar input conditions contributed to it.

- ☐ Anomaly Detection
- ☐ Outlier Detection
- ☐ Out-of-Distribution Detection
- ☐ Distribution Shift
- ☐ Multimodal OOD
- ☐ Graph OOD
- ☐ Benchmark Design

Tools from this group have been cited in a U.S. Senate committee report, a NIST special publication, and the U.S. Department of Defense Generative AI Toolkit. PyOD was selected by the European Space Agency for spacecraft anomaly detection and is the subject of five published books. TrustLLM serves as an official benchmark in all four editions of the Future of Life Institute AI Safety Index through Summer 2026.

PROFESSIONAL EXPERIENCE **University of Southern California**
Thomas Lord Department of Computer Science
 Assistant Professor (Tenure-Track) Aug. 2023 - Present

- **F**oundations **O**f **R**obust **T**rustworthy **I**ntelligent **S**ystems (**FORTIS**) Lab: [Link](#)
- USC Machine Learning Center (MaSCle): [Link](#)

Fig Work

AI-powered marketplace for early-career talent
 Chief Scientific Advisor Jun. 2026 - Present

PwC Canada

Consulting & Deals
 Senior Consultant (Data Scientist) Aug. 2017 - Jun. 2019
 Consultant (Data Scientist) Feb. 2017 - Jul. 2017
 Research Associate (Intern) May. 2016 - Jan. 2017

EDUCATION **Carnegie Mellon University** Pittsburgh, PA
Ph.D. in Information Systems and Management Sep. 2019 - May. 2023
 Heinz College of Information Systems and Public Policy

- **Affiliation:** CMU automated learning systems group (Catalyst) and Data Analytics Techniques Algorithms (DATA) Lab
- **Advisors and Mentors:** CMU: Prof. Leman Akoglu, Prof. Zhihao Jia, and Prof. George Chen. I collaborate with Prof. Jure Leskovec at Stanford, and Prof. Philip S. Yu at UIC.

University of Toronto Toronto, ON
Master of Science in Computer Science Sep. 2015 - Dec. 2016

University of Cincinnati Cincinnati, OH
Bachelor of Science in Computer Engineering Sep. 2010 - May. 2015
Minor: Computer Science and Mathematics

🏆 AWARDS,
GRANTS, AND
FUNDING

As Principal Investigator (August 2023 onwards)

NSF FIRE	<i>Funding</i>	Aug. 2026
Meta	<i>Sponsored Research</i>	Aug. 2026
Foresight Institute AI for Safety & Science Nodes	<i>Gift</i>	Apr. 2026
NVIDIA Academic Grant Program	<i>Gift</i>	Mar. 2026
Amazon Research Awards, Fall 2025	<i>Gift</i>	Mar. 2026
USC CCSL Seed Funding	<i>Small Grant</i>	Jan. 2026
Second Prize CCC Award @ IEEE ICDM, BlueSky Track	<i>Recognition</i>	Nov. 2025
Best Short Paper Award @ ACM SIGSPATIAL	<i>Recognition</i>	Nov. 2025
NSF Pathways to Enable Open-Source Ecosystems (POSE), Phase I	<i>Funding</i>	Aug. 2025
Capital One Research Awards	<i>Grant</i>	Oct. 2024
Amazon Research Awards, Spring 2024	<i>Gift</i>	Aug. 2024
Best Paper Award @ KDD Resource-Efficient Learning Workshop	<i>Recognition</i>	Aug. 2024
NSF ATD	<i>Funding</i>	Aug. 2024
NSF POSE Phase II	<i>Funding</i>	Jun. 2024
Google Cloud Research Innovators	<i>Recognition</i>	Mar. 2024
AAAI New Faculty Highlights	<i>Recognition</i>	Feb. 2024

*Note: Monetary values represent **my portion** of the funding. Total project budgets may be larger.*

Prior to Principal Investigator Role (Before August 2023)

Meta 2022 AI4AI Research Award (student co-PI)	<i>Recognition</i>	Oct. 2022
The Norton Labs Graduate Fellowship	<i>Fellowship</i>	Mar. 2022
CMU Presidential Fellowship	<i>Fellowship</i>	2019
Mitacs-Accelerate Research and Development Funding	<i>Funding</i>	2016-2017
University Global Award and Scholarship	<i>Scholarship</i>	2010-2015
Mantei/Mae Award & Scholar	<i>Award</i>	2012-2015
Engineer of the Month	<i>Recognition</i>	Jun. 2014

Note: Monetary values are omitted for awards and recognitions received prior to PI role.

SELECTED
OPEN-SOURCE
SOFTWARE,
BENCHMARKS,
AND SYSTEMS

Representative open-source research artifacts spanning anomaly detection, trustworthy AI, agent security, and research tooling.

Primary Contributions

- **PyOD** (*Library*; updated Oct. 2026; **10.0K stars**) PyOD 3 is the most comprehensive Python library for anomaly detection: 60+ detectors across tabular, time series, graph, text, and image data, with an ADEngine orchestration core and an agentic investigation layer driven through natural language. 59M+ downloads. Links: docs — paper.
- **Anomaly-Detection-Resources** (*Resource Hub*; updated Mar. 2026; **9.4K stars**) A curated resource hub for anomaly detection, including papers, benchmarks, tutorials, datasets, tools, and community references.
- **CS-Paper-Checklist** (*Research Toolkit*; updated May. 2026; **1.6K stars**) A practical sanity checklist for improving CS paper quality, structure, and submission readiness.
- **PyGOD** (*Library*; updated Nov. 2024; **1.5K stars**) A Python library for graph outlier detection, with unified APIs and broad coverage of graph anomaly detection algorithms. Links: docs — paper.
- **ADBench** (*Benchmark*; updated Jan. 2026; **1.0K stars**) Official benchmark for anomaly detection with large-scale evaluations across unsupervised, semi-supervised, and supervised settings. Links: paper.
- **agent-style** (*Developer Toolkit*; updated Sep. 2026; **706 stars**) A drop-in rules package that shapes how AI coding and writing agents produce prose at generation time, using 21 curated English style rules (12 from canonical writing authorities; 9 from field observation of LLM output). Works with Claude Code, Codex, Copilot, Cursor, Aider, and other AGENTS.md-compliant tools. Links: docs — demo.
- **combo** (*Library*; updated Sep. 2026; **659 stars**) A Python toolbox for machine learning model and score combination across classification, clustering, and anomaly detection tasks. Links: docs — paper.

- **Aegis** (*Security / Compliance Platform*; updated Sep. 2026; [486 stars](#)) The open-source firewall for AI agents, with pre-execution policy checks, human-in-the-loop approvals, and tamper-evident audit trails for safer and more accountable deployment. Links: [docs](#) — [paper](#) — [demo](#).
- **SUOD** (*Library / System*; updated Sep. 2026; [395 stars](#)) An acceleration framework for large-scale unsupervised heterogeneous outlier detection with efficient training and inference. Links: [docs](#) — [paper](#).
- **anywhere-agents** (*Developer Toolkit*; updated Oct. 2026; [248 stars](#)) One config to rule all your AI agents: portable across every project and session, effective through curated writing, routing, and skills, and safer via a PreToolUse guard that blocks destructive Git and GitHub commands. Supports Claude Code and Codex today, with plans to grow. Links: [docs](#).
- **agent-audit** (*Security Toolkit*; updated Sep. 2026; [237 stars](#)) An open-source tool for auditing AI agent code and configurations, with OWASP Agentic Top 10 coverage, taint analysis, and MCP configuration auditing. Links: [docs](#) — [paper](#) — [demo](#).
- **awesome-auditable-ai** (*Resource Hub*; updated Sep. 2026; [149 stars](#)) A curated list of papers, tools, datasets, benchmarks, and standards for building, evaluating, and auditing reliable AI agents, organized around a reader-first reliability map.
- **AD-AGENT** (*Platform*; updated Apr. 2026; [104 stars](#)) An LLM-driven multi-agent anomaly detection platform for end-to-end workflows across tabular, graph, and time-series settings. Links: [docs](#) — [paper](#).
- **AD-LLM** (*Benchmark*; updated Oct. 2025; [44 stars](#)) A benchmark for using large language models in anomaly detection, covering zero-shot detection, data augmentation, and model selection. Links: [paper](#).
- **auditable** (*Agent Auditability*; updated Sep. 2026; [27 stars](#)) An open-source system of record for AI-agent decisions: it captures what each decision relied on, replays it against live state and rolls back the committed action when it no longer holds, and ranks the steps of a finished run by structural blast share, on one typed graph across an agent’s plan, run, and review. Links: [docs](#) — [paper](#) — [demo](#).
- **NLP-ADBench** (*Benchmark*; updated Oct. 2025; [21 stars](#)) A benchmark for anomaly detection in NLP, including NLPAD datasets and standardized evaluation across end-to-end and embedding-based methods. Links: [paper](#).
- **CatchBench** (*Benchmark*; updated Sep. 2026; [11 stars](#)) A benchmark for auditing agent failures across the full lifecycle. It scores one auditor’s question against three information states: the declared configuration before a run, a growing prefix of the trace, and the finished trace. The release covers 72 entrants and 1187 declared configurations. Contrasts that do not separate are published unresolved rather than ranked. Links: [paper](#).

Selected Collaborations

- **TODS** (*System / AutoML*; updated Sep. 2023; [1.7K stars](#)) An automated time-series outlier detection system with end-to-end pipeline support for preprocessing, feature extraction, model search, and detection. Links: [docs](#) — [paper](#).
- **TDC** (*Benchmark / Platform*; updated Jul. 2025; [1.3K stars](#)) Therapeutics Data Commons, a platform for AI-ready therapeutic datasets, benchmarks, and evaluation workflows across drug discovery tasks. Links: [docs](#) — [paper](#).
- **TrustLLM** (*Benchmark / Toolkit*; updated Oct. 2026; [633 stars](#)) A trustworthiness benchmark and toolkit for LLMs; collaboration included follow-on NSF-supported development with the core team. Links: [docs](#) — [paper](#).
- **TrustEval-toolkit** (*Benchmark / Toolkit*; updated Aug. 2025; [137 stars](#)) A toolkit and benchmark for evaluating trustworthiness of generative foundation models across LLM, VLM, and text-to-image settings. Links: [docs](#) — [paper](#).
- **LangSkills** (*Toolkit / Knowledge Base*; updated Mar. 2026; [116 stars](#)) An evidence-backed skill library and pipeline for AI agents, with offline searchable bundles distilled from papers and technical sources.

129. Jinyi Ye[✉], Yuangang Li, Chenxiao Yu, Preyashi Poddar, Priyanka Dey, Longtian Ye, Zihan Wang, Xiyang Hu, Emilio Ferrara[✉], [Yue Zhao[✉]](#)
MetaPersona: Task-Grounded Synthetic Populations from Empirical Social Science
arXiv preprint arXiv:2609.38392
128. Jesson Wang[†], Shawn Li[†], Wei Yang, Franck Dernoncourt, Ryan A. Rossi, Charith Peris, [Yue Zhao](#)
Controlled Decoding Attacks on Black-Box LLMs
arXiv preprint arXiv:2609.36956
127. Prince Zizhuang Wang^{†✉}, Chenhao Liang[†], Zelong Xu[†], Aojie Yuan[†], Xiaolin Zhou[†], Haiyue Zhang, [Yue Zhao](#), Xiyang Hu, Shuli Jiang
CUA-SWE: When Computer-Use Agents Meet Visual Software Engineering
arXiv preprint arXiv:2609.32600
126. [Yue Zhao](#), Huayue Gu, Yushun Dong, Xiyang Hu
QEMScore: How Much Does the Measurement Add to Learned Quantum Error Mitigation?
arXiv preprint arXiv:2609.17896
125. [Yue Zhao](#), Mengyuan Li, Ruolin Li, Prince Zizhuang Wang, Shuli Jiang, Linsey Pang, Xiongye Xiao, Xiyang Hu
CatchBench: When Can an Agent Failure Be Caught?
arXiv preprint arXiv:2608.22808
124. Shawn Li, Wei Yang, Jake Zhong, Jiate Li, Jiawei Yang, You Qin, Ryan Rossi, Franck Dernoncourt, Roger Zimmermann, Yue Wang, Zhengzhong Tu, Vicente Ordonez, Mohit Bansal, [Yue Zhao](#)
JigShape: Evaluating Visual-Geometric Reasoning in VLMs through Jigsaw Puzzles
arXiv preprint arXiv:2607.27670
123. Yue Huang, Wenjie Wang, Han Bao, Yuchen Ma, Xiaonan Luo, Yi Nian, Haomin Zhuang, Zheyuan Liu, [Yue Zhao](#), Xiangliang Zhang
MemoHarness: Agent Harnesses That Learn from Experience
arXiv preprint arXiv:2607.14159
122. Tianming Sha, [Yue Zhao](#), Lichao Sun, Yushun Dong
SkillCenter: A Large-Scale Source-Grounded Skill Library for Autonomous AI Agents
arXiv preprint arXiv:2607.07676
121. Shawn Li, [Yue Zhao](#)
Agent Safety Is Action Alignment
arXiv preprint arXiv:2606.28739
120. [Yue Zhao](#)
GRADE: Graph Representation of LLM Agent Dependency and Execution
arXiv preprint arXiv:2606.22741
119. Ojas Nimase, Jiate Li, [Yue Zhao](#), Yushun Dong
Can Subgraph Explanations Be Weaponized to Steal Graph Neural Networks?
arXiv preprint arXiv:2605.30470
118. Ojas Nimase, Zhe Chen, Gengpei Qi, [Yue Zhao](#), Xiyang Hu
GEO-Bench: Benchmarking Ranking Manipulation in Generative Engine Optimization
arXiv preprint arXiv:2605.29107
117. Shawn Li, Chenxiao Yu, Han Wang, Wei Yang, Ryan Rossi, Franck Dernoncourt, Xiyang Hu, Philip Yu, Chaowei Xiao, Huan Zhang, [Yue Zhao](#)
FORTIS: Benchmarking Over-Privilege in Agent Skills
arXiv preprint arXiv:2605.09163
116. Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan Rossi, Kaize Ding, [Yue Zhao](#)
No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents
arXiv preprint arXiv:2604.01350
115. Haiyue Zhang, Yi Nian, [Yue Zhao](#)
Agent Audit: A Security Analysis System for LLM Agent Applications
arXiv preprint arXiv:2603.22853
114. Yi Nian, Haosen Cao, Shenzhe Zhu, Henry Peng Zou, Qingqing Luan, Yudi Zhang, [Yue Zhao](#)
When Only the Final Text Survives: Implicit Execution Tracing for Multi-Agent Auditing
Under submission
arXiv preprint arXiv:2603.17445

113. Aojie Yuan, Zhiyuan Su, Yue Zhao
AEGIS: No Tool Call Left Unchecked – A Pre-Execution Firewall and Audit Layer for AI Agents
Under submission
arXiv preprint arXiv:2603.12621
112. Rujie Ye, Jiayi Zhang, Zhuoxin Liu, Zihao Zhu, Siyuan Yang, Li Li, Tianfu Fu, Franck Dernoncourt, Yue Zhao, Jiacheng Zhu, Ryan Rossi, Wenhao Chai, Zhengzhong Tu
Agent Banana: High-Fidelity Image Editing with Agentic Thinking and Tooling
Under submission
arXiv preprint arXiv:2602.09084
111. Xiaolin Zhou, Zheng Luo, Yicheng Gao, Qixuan Chen, Xiyang Hu, Yue Zhao, Ruishan Liu
Fairness or Fluency? An Investigation into Language Bias of Pairwise LLM-as-a-Judge
Under submission
arXiv preprint arXiv:2601.13649
110. Kay Liu, Yuwei Han, Haoyan Xu, Henry Peng Zou, Yue Zhao, Philip S. Yu
TAGFN: A Text-Attributed Graph Dataset for Fake News Detection in the Age of LLMs
Under submission
arXiv preprint arXiv:2511.21624
109. Haoyan Xu, Ruizhi Qian, Zhengtao Yao, Ziyi Liu, Li Li, Yuqi Li, Yanshu Li, Wenqing Zheng, Daniele Rosa, Daniel Barcklow, Senthil Kumar, Jieyu Zhao, Yue Zhao
LLM-Powered Text-Attributed Graph Anomaly Detection via Retrieval-Augmented Reasoning
Under submission
arXiv preprint arXiv:2511.17584
108. Haoyan Xu, Ruizhi Qian, Jiate Li, Yushun Dong, Minghao Lin, Hanson Yan, Zhengtao Yao, Qinghua Liu, Junhao Dong, Ruopeng Huang, Yue Zhao✶, Mengyuan Li✶
A Systematic Study of Model Extraction Attacks on Graph Foundation Models
Under submission
arXiv preprint arXiv:2511.11912
107. Yuexing Hao, Yue Huang, Haoran Zhang, Chenyang Zhao, Zhenwen Liang, Paul Pu Liang, Yue Zhao, Lichao Sun, Saleh Kalantari, Xiangliang Zhang, Marzyeh Ghassemi
The Role of Computing Resources in Publishing Foundation Model Research
Under submission
arXiv preprint arXiv:2510.13621
106. Wang Wei, Tiankai Yang, Hongjie Chen, Yue Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry
Learning to Route LLMs from Bandit Feedback: One Policy, Many Trade-offs
Under submission
arXiv preprint arXiv:2510.07429
105. Langzhou He, Junyou Zhu, Fangxin Wang, Junhua Liu, Haoyan Xu, Yue Zhao, Philip S. Yu, Qitian Wu
Can Molecular Foundation Models Know What They Don't Know? A Simple Remedy with Preference Optimization
Under submission
arXiv preprint arXiv:2509.25509
104. Yuehan Qin, Li Li, Defu Cao, Tiankai Yang, Yue Zhao
M3OOD: Automatic Selection of Multimodal OOD Detectors
Under submission
arXiv preprint arXiv:2508.11936
103. Haoyan Xu, Zhengtao Yao, Xuzhi Zhang, Ziyi Wang, Langzhou He, Yushun Dong, Philip S. Yu, Mengyuan Li, Yue Zhao
GLIP-OOD: Zero-Shot Graph OOD Detection with Foundation Model
Under submission
arXiv preprint arXiv:2504.21186
102. Haoyan Xu, Zhengtao Yao, Ziyi Wang, Zhan Cheng, Xiyang Hu, Mengyuan Li, Yue Zhao
Graph Synthetic Out-of-Distribution Exposure with Large Language Models
Under submission
arXiv preprint arXiv:2504.21198
101. Kaixiang Zhao, Lincan Li, Kaize Ding, Neil Zhenqiang Gong, Yue Zhao, Yushun Dong
A Survey of Model Extraction Attacks and Defenses in Distributed Computing Environments

Under submission
arXiv preprint arXiv:2502.16065

100. Shixuan Li, Wei Yang, Peiyu Zhang, Xiongye Xiao, Defu Cao, Yuehan Qin, Xiaole Zhang, [Yue Zhao](#), Paul Bogdan
ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models

Under submission
arXiv preprint arXiv:2502.11059

99. Lincan Li, Jiaqi Li, Catherine Chen[✉], Fred Gui[✉], other collaborators, [Yue Zhao](#)[✉], Yushun Dong[✉]
Political-LLM: Large Language Models in Political Science

Under submission
arXiv preprint arXiv:2412.06864

98. Chenxiao Yu, Jinyi Ye, Yuangang Li, Zhaotian Weng, Zheng Li, Emilio Ferrara, Xiyang Hu[✉], [Yue Zhao](#)[✉]
A Large-Scale Simulation on Large Language Models for Decision-Making in Political Science

Under submission
arXiv preprint arXiv:2412.15291

97. Junda Wu, Hanjia Lyu, Yu Xia, Zhehao Zhang, Joe Barrow, Ishita Kumar, Mehnoosh Mirtahebi, Hongjie Chen, Ryan A. Rossi, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, Jiuxiang Gu, Nesreen K. Ahmed, Yu Wang, Xiang Chen, Hanieh Deilamsalehy, Namyong Park, Sungchul Kim, Huanrui Yang, Subrata Mitra, Zhengmian Hu, Nedim Lipka, [Yue Zhao](#), Jiebo Luo, Julian McAuley
Personalized Multimodal Large Language Models: A Survey

Under submission
arXiv preprint arXiv:2412.02142

Peer-reviewed Journal Papers

96. Jiechao Gao[†], Yuangang Li[†], Jie Wang, Michael Lepech, [Yue Zhao](#), Brad Campbell
H-FedSN: Personalized Sparse Networks for Efficient and Accurate Hierarchical Federated Learning for IoT Applications
npj Artificial Intelligence, 2026.
95. Yuehan Qin, Shawn Li, Yi Nian, Xinyan Velocity Yu, [Yue Zhao](#)[✉], Xuezhe Ma[✉]
Don't Let It Hallucinate: Premise Verification via Retrieval-Augmented Logical Reasoning
Transactions on Machine Learning Research (TMLR), 2026
94. Haoyan Xu, Kay Liu, Zhengtao Yao, Philip S. Yu, Mengyuan Li, Kaize Ding[✉], [Yue Zhao](#)[✉]
LEGO-Learn: Label-Efficient Graph Open-Set Learning
Transactions on Machine Learning Research (TMLR), 2025
93. Hao Dong, Gaetan Frusque, [Yue Zhao](#), Eleni Chatzi, Olga Fink
NNG-Mix: Improving Semi-supervised Anomaly Detection with Pseudo-anomaly Generation
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024
92. Ling Yang[†], Zhilong Zhang[†], Yang Song, Shenda Hong, Runsheng Xu, [Yue Zhao](#), Wentao Zhang, Bin Cui, Ming-Hsuan Yang
Diffusion Models: A Comprehensive Survey of Methods and Applications
ACM Computing Surveys (CSUR), 2023
91. [Yue Zhao](#)[†], Martin Q. Ma[†], Xiaorong Zhang, Leman Akoglu
The Need for Unsupervised Outlier Model Selection: A Review and Evaluation of Internal Evaluation Strategies
ACM SIGKDD Explorations Newsletter (SIGKDD Explor.), 2023
90. Kexin Huang[†], Tianfan Fu[†], Wenhao Gao[†], [Yue Zhao](#), Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Artificial Intelligence Foundation for Therapeutic Science
Nature Chemical Biology (NCHEMB), 2022
89. Zheng Li[†], [Yue Zhao](#)[†], Xiyang Hu, Nicola Botta, Cezar Ionescu, George H. Chen
ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2022.
88. [Yue Zhao](#), Zain Nasrullah, Zheng Li
PyOD: A Python Toolbox for Scalable Outlier Detection
Journal of Machine Learning Research (JMLR), 2019.

Conference & Workshop Papers

87. Wei Yang, Li Li, Yuehan Qin, Yawei Wang, Mingxi Wang, Shixuan Li, Tiankai Yang, Jiate Li, Jesse Thomason, Xuezhe Ma, [Yue Zhao](#)
SynCo: Data Synthesis Co-Training for Self-Evolving LLMs via Multi-Agent Reinforcement Learning
NeurIPS 2026 Workshop on Towards Test-Time Continual Learning Agents (TTCL), Poster.
86. Tiankai Yang, Yi Nian, Xinyuan Li, Ruiyao Xu, Henry Peng Zou, Kaize Ding, Xiyang Hu, Yan Liu, [Yue Zhao](#)
Cat-DPO: Category-Adaptive Safety Alignment
Advances in Neural Information Processing Systems (NeurIPS), 2026.
85. Shawn Li, You Qin, Jiate Li, Charith Peris, Lisa Bauer, Roger Zimmermann, [Yue Zhao](#)
Geometry over Density: Few-Shot Cross-Domain OOD Detection
Advances in Neural Information Processing Systems (NeurIPS), 2026.
84. Yuehan Qin, Li Li, Linxin Song, Wei Yang, Jiate Li, Yuqing Yang, [Yue Zhao](#)
Memory Retrieval for Changing Preferences
Advances in Neural Information Processing Systems (NeurIPS), 2026.
83. Chenxiao Yu, Bowen Yi, Farzan Karimi-Malekabadi, Suhaib Abdurahman, Jinyi Ye, Shrikanth Narayanan, [Yue Zhao](#), Morteza Dehghani
Tracing Moral Foundations in Large Language Models
Advances in Neural Information Processing Systems (NeurIPS), 2026.
82. Yingzi Ma, Zhengyue Zhao, Xiaogeng Liu, Minhui Xue, [Yue Zhao](#), Chaowei Xiao
MaskForge: Structure-Aware Adaptive Attacks for Jailbreaking Diffusion Large Language Models
Advances in Neural Information Processing Systems (NeurIPS), 2026.
81. Yicheng Gao, Xiaolin Zhou, Yahan Li, [Yue Zhao](#), Ruishan Liu
MedExAgent: Training LLM Agents to Ask, Examine, and Diagnose in Noisy Clinical Environments
Advances in Neural Information Processing Systems (NeurIPS), 2026.
80. Haolin Chen, Deon Metelski, Leon Qi, Tao Xia, Joonyul Lee, Steve Brown, Kevin Riley, Frank Wang, T. Y. Alvin Liu, Hank Capps MD, Zeyu Tang, Xiangchen Song, Lingjing Kong, Fan Feng, Tianyi Zeng, Zhiwei Liu, Zixian Ma, Hang Jiang, Fangli Geng, Yuan Yuan, Chenyu You, Qingsong Wen, Hua Wei, Yanjie Fu, [Yue Zhao](#), Carl Yang, Biwei Huang, Kun Zhang, Caiming Xiong, Sanmi Koyejo, Eric P. Xing, Philip S. Yu, Weiran Yao
CHI-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?
Advances in Neural Information Processing Systems (NeurIPS Evaluations and Datasets Track), 2026.
79. Shawn Li, [Yue Zhao](#)
The Autonomy Tax: Defense Training Breaks LLM Agents
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
78. Wenjie Wang, Yue Huang, Zhengqing Yuan, Han Bao, Shiyi Du, Yuchen Ma, [Yue Zhao](#), Yanfang Ye, Xiangliang Zhang
SpecAlign: Efficient Specification-Grounded Alignment of Large Language Models via Synthetic Data
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
77. Yi Nian, Tiankai Yang, Yudi Zhang, Qi Pan, Zelong Xu, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, [Yue Zhao](#)
DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment
Findings of the Association for Computational Linguistics: EMNLP, 2026.
76. Prince Zizhuang Wang, Aojie Yuan, Haiyue Zhang, Xiyang Hu, [Yue Zhao](#), Shuli Jiang
WeClawArena: An Auditable Sandbox and Benchmark for Cross-User Agents Collaboration and Security in Human-Centered Agent Networks
Findings of the Association for Computational Linguistics: EMNLP, 2026.
75. Zixiang Xu, Yanbo Wang, Yue Huang, Haomin Zhuang, Yujun Zhou, Jiayi Ye, Sixian Li, Zirui Song, Lang Gao, Chenxi Wang, Zhaorun Chen, Wang Pan, [Yue Zhao](#), Jieyu Zhao, Xiangliang Zhang, Xiuying Chen
SocialMaze: A Benchmark for Evaluating and Enhancing Social Reasoning in Large Language Models in Complex Social Environments
Findings of the Association for Computational Linguistics: EMNLP, 2026. Earlier versions presented at *SocialSim @ COLM 2025 (Spotlight)* and *ResponsibleFM @ NeurIPS 2025*.

74. Ojas Nimase, Zhengao Li, Xinyu Wei, Jinglin Hu, LuoJia Liu, Kangyi Zhao, [Yue Zhao](#), Yushun Dong
FinanceLLM: A Survey of Large Language Models in Finance
Findings of the Association for Computational Linguistics: EMNLP, 2026.
73. Aojie Yuan, Yi Nian, Haiyue Zhang, Zijian Su, [Yue Zhao](#)
SEVA: Self-Evolving Verification Agent with Process Reward for Fact Attribution
ICML 2026 Workshops (AI4GOOD; FAGEN).
72. Wang Wei, Hongzheng Yang, Tiankai Yang, Samyadeep Basu, Hongjie Chen, [Yue Zhao](#), Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry
FlexRouter: Learning Complementary Model Sets for Flexible LLM Routing
Conference on Language Modeling (COLM), 2026.
71. Yiming Tang, Yi Fan, Chenxiao Yu, Tiankai Yang, [Yue Zhao](#), Xiyang Hu
StealthRank: LLM Ranking Manipulation via Stealthy Prompt Optimization
ICML Workshop on Trustworthy AI for Good (AI4GOOD), 2026.
70. Han Bao, Yue Huang, Yanbo Wang, Jiayi Ye, Xiangqi Wang, Xiuying Chen, [Yue Zhao](#), Tianyi Zhou, Mohamed Elhoseiny, Xiangliang Zhang
AutoDavis: Automatic and Dynamic Evaluation Protocol of Large Vision-Language Models on Visual Question-Answering
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD Datasets and Benchmarks Track), 2026.
69. Zhisheng Qi, Utkarsh Sahu, Li Ma, Haoyu Han, Ryan Rossi, Franck Dernoncourt, Mahantesh Halappanavar, Nesreen Ahmed, Yushun Dong, [Yue Zhao](#), Yu Zhang, Yu Wang
Benchmarking Knowledge-Extraction Attack and Defense on Retrieval-Augmented Generation
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD Datasets and Benchmarks Track),  **Oral**, 2026.
68. Jiate Li, Defu Cao, Li Li, Wei Yang, Yuehan Qin, Chenxiao Yu, Tiannuo Yang, Ryan A. Rossi, Yan Liu, Xiyang Hu, [Yue Zhao](#)
“Someone Hid It”: Query-Agnostic Black-Box Attacks on LLM-Based Retrieval
International Conference on Machine Learning (ICML), 2026.
67. Shawn Li, Chenxiao Yu, Zhiyu Ni, Hao Li, Charith Peris, Chaowei Xiao, [Yue Zhao](#)
Defenses Against Prompt Attacks Learn Surface Heuristics
Annual Meeting of the Association for Computational Linguistics (ACL), 2026.
66. Ruiyao Xu, Mihir Parmar, Tiankai Yang, Zhengyu Hu, [Yue Zhao](#), Kaize Ding
CoAct: Co-Active LLM Preference Learning with Human-AI Synergy
Annual Meeting of the Association for Computational Linguistics (ACL),  **Oral**, 2026.
65. Jinbo Liu, Defu Cao, Yifei Wei, Tianyao Su, Yuan Liang, Yushun Dong, Yan Liu, [Yue Zhao](#), Xiyang Hu
Topology Matters: Measuring Memory Leakage in Multi-Agent LLMs
Findings of the Association for Computational Linguistics: ACL, 2026.
64. Yi Nian, Aojie Yuan, Haiyue Zhang, Jiate Li, Li Li, Xiyang Hu, Hua Wei, Xiongye Xiao, Chaowei Xiao, [Yue Zhao](#)
Auditable Agents
ACM AI Leadership Summit, 2026 (condensed 4-page version). Earlier version presented at the *ACL 2026 KnowFM Workshop*.
arXiv preprint [arXiv:2604.05485](#)
63. Yixuan Du, Chenxiao Yu, Haoyan Xu, Ziyi Wang, [Yue Zhao](#), Xiyang Hu
Multimodal Generative Engine Optimization: Rank Manipulation for Vision-Language Model Rankers
ACL Workshop on Towards Knowledgeable Foundation Models (KnowFM), 2026.
arXiv preprint [arXiv:2601.12263](#)
62. Shawn Li, Ryan Rossi, Sungchul Kim, Sunav Choudhary, Franck Dernoncourt, Puneet Mathur, Zhengzhong Tu, [Yue Zhao](#)
Charts Are Not Images: On the Challenges of Scientific Chart Editing
International Conference on Learning Representations (ICLR), 2026
61. Weidi Luo, Tianyu Lu, Qiming Zhang, Xiaogeng Liu, Bin Hu, [Yue Zhao](#), Jieyu Zhao, Song Gao, Patrick McDaniel, Zhen Xiang, Chaowei Xiao
Doxing via the Lens: Revealing Privacy Leakage in Image Geolocation for Agentic Multi-Modal Large Reasoning Model
International Conference on Learning Representations (ICLR), 2026

60. Yue Huang, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Yujun Zhou, Yanbo Wang, Jiayi Ye, Jiawen Shi, Qihui Zhang, Yuan Li, Han Bao, Zhaoyi Liu, Tianrui Guan, Dongping Chen, Ruoxi Chen, other authors, Yue Zhao, other authors, Xiangliang Zhang
On the Trustworthiness of Generative Foundation Models: Guideline, Assessment, and Perspective
International Conference on Learning Representations (ICLR), 2026
<https://trustgen.github.io/>
59. Chengxuan Qian, Shuo Xing, Shawn Li, Yue Zhao, Zhengzhong Tu
DecAlign: Hierarchical Cross-Modal Alignment for Decoupled Multimodal Representation Learning
International Conference on Learning Representations (ICLR), 2026
58. Xiong Xiao Xu, Haoran Wang, Yueqing Liang, Philip S. Yu, Yue Zhao, Kai Shu
Can Multimodal LLMs Perform Time Series Anomaly Detection?
The Web Conference (WWW), 2026
57. Bo Ni, Leyao Wang, Yu Wang, Branislav Kveton, Franck Dernoncourt, Yu Xia, Hongjie Chen, Reuben Leura, Samyadeep Basu, Subhojyoti Mukherjee, Puneet Mathur, Nesreen Ahmed, Junda Wu, Li Li, Huixin Zhang, Ruiyi Zhang, Tong Yu, Sungchul Kim, Jiuxiang Gu, Zhengzhong Tu, Alexa Siu, Zichao Wang, David Seunghyun Yoon, Nedim Lipka, Namyong Park, Zihao Lin, Trung Bui, Yue Zhao, Tyler Derr, Ryan A. Rossi
A Survey on LLM-based Conversational User Simulation
Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026
56. Yuangang Li, Yiqing Shen, Yi Nian, Jiechao Gao, Ziyi Wang, Chenxiao Yu, Li Li, Jie Wang, Xiyang Hu, Yue Zhao
Mitigating Hallucinations in Large Language Models via Causal Reasoning
Fortieth AAAI Conference on Artificial Intelligence (AAAI), 2026
55. Ojas Nimase, Yue Zhao, Yushun Dong
Navigating Between Explainability and Extractability in Machine Learning as a Service
IEEE International Conference on Data Mining (ICDM) BlueSky Track,  **Second Prize CCC Award**, 2025.
54. Tiankai Yang, Junjun Liu, Wingchun Siu, Jiahang Wang, Zhuangzhuang Qian, Chanjuan Song, Cheng Cheng, Xiyang Hu, Yue Zhao
AD-AGENT: A Multi-agent Framework for End-to-end Anomaly Detection
Findings of the Association for Computational Linguistics: IJCNLP-AAACL, 2025.
53. Ruosi Shao, Md Shamim Seraj, Kangyi Zhao, Yingtao Luo, Lincan Li, Bolin Shen, Averi Bates, Yue Zhao, Chongle Pan, Lisa Hightow-Weidman, Shayok Chakraborty, Yushun Dong
LLM-Empowered Patient-Provider Communication: A Data-Centric Survey From a Clinical Perspective
Findings of the Association for Computational Linguistics: IJCNLP-AAACL, 2025.
52. Shawn Li, Peilin Cai, Ryan A. Rossi, Franck Dernoncourt, Branislav Kveton, Junda Wu, Tong Yu, Linxin Song, Tiankai Yang, Yuehan Qin, Nesreen K. Ahmed, Samyadeep Basu, Subhojyoti Mukherjee, Ruiyi Zhang, Zhengmian Hu, Bo Ni, Yuxiao Zhou, Zichao Wang, Yue Huang, Yu Wang, Xiangliang Zhang, Philip S. Yu, Xiyang Hu, Yue Zhao
A Personalized Conversational Benchmark: Towards Simulating Personalized Conversations
NeurIPS Workshop on Multi-Turn Interactions in Large Language Models (MTI-LLM),  **Spotlight**, 2025.
arXiv preprint arXiv:2505.14106
51. Yanbo Wang, Zixiang Xu, Yue Huang, Xiangqi Wang, Zirui Song, Lang Gao, Chenxi Wang, Xiangru Tang, Yue Zhao, Arman Cohan, Xiangliang Zhang, Xiuying Chen
DyFlow: Dynamic Workflow Framework for Agentic Reasoning
Advances in Neural Information Processing Systems (NeurIPS), 2025
50. Shawn Li, Jiashu Qu, Linxin Song, Yuxiao Zhou, Yuehan Qin, Tiankai Yang, Yue Zhao
Treble Counterfactual VLMs: A Causal Approach to Hallucination
Findings of the Association for Computational Linguistics: EMNLP, 2025.
49. Yuangang Li, Jiaqi Li, Zhuo Xiao, Tiankai Yang, Yi Nian, Xiyang Hu, Yue Zhao
NLP-ADBench: NLP Anomaly Detection Benchmark
Findings of the Association for Computational Linguistics: EMNLP, 2025.
48. Lincan Li, Eren Erman Ozguven, Yue Zhao, Guang Wang, Yiqun Xie, Yushun Dong
TyphoFormer: Language-Augmented Transformer for Accurate Typhoon Track Forecasting
ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL),  **Best Short Paper Award**, 2025.

47. Bolin Shen, Eren Erman Ozguven, [Yue Zhao](#), Guang Wang, Yiqun Xie, Yushun Dong
Learning from the Storm: A Multivariate Machine Learning Approach to Predicting Hurricane-Induced Economic Losses
ACM SIGSPATIAL International Workshop on Spatial Intelligence for Smart and Connected Communities (SpatialConnect), 2025
46. Yi Nian[†], Shenzhe Zhu[†], Yuehan Qin, Shawn Li, Ziyi Wang, Chaowei Xiao, [Yue Zhao](#)
JailDAM: Jailbreak Detection with Adaptive Memory for Vision-Language Model
Conference on Language Modeling (COLM), 2025.
45. Shawn Li, Peilin Cai, Yuxiao Zhou, Zhiyu Ni, Renjie Liang, You Qin, Yi Nian, Zhengzhong Tu, Xiyang Hu, [Yue Zhao](#)
Secure On-Device Video OOD Detection Without Backpropagation
International Conference on Computer Vision (ICCV), 2025.
44. Zerui Xu, Fang Wu, Yingzhou Lu, Yuanyuan Zhang, [Yue Zhao](#)
Retrieval-Reasoning Large Language Model-based Synthetic Clinical Trial Generation
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD Workshop on AI Agent for Information Retrieval), 2025.
ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB), 2025.
43. Haoyan Xu[†], Zhengtao Yao[†], Yushun Dong, Ziyi Wang, Ryan A. Rossi, Mengyuan Li, [Yue Zhao](#)
Few-Shot Graph Out-of-Distribution Detection with LLMs
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD), 2025.
42. Tiankai Yang[†], Yi Nian[†], Shawn Li, Ruiyao Xu, Yuangang Li, Jiaqi Li, Zhuo Xiao, Xiyang Hu, Ryan Rossi, Kaize Ding, Xia Hu, [Yue Zhao](#)
AD-LLM: Benchmarking Large Language Models for Anomaly Detection
Findings of the Association for Computational Linguistics (ACL Findings), 2025.
41. Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, Hanjia Lyu, Joe Barrow, Hongjie Chen, Franck Dernoncourt, Branislav Kveton, Tong Yu, Ruiyi Zhang, Jiuxiang Gu, Nesreen K. Ahmed, Yu Wang, Xiang Chen, Hanieh Deilamsalehy, Sungchul Kim, Zhengmian Hu, [Yue Zhao](#), Nedim Lipka, Seunghyun Yoon, Ting-Hao Kenneth Huang, Zichao Wang, Puneet Mathur, Soumyabrata Pal, Koyel Mukherjee, Zehao Zhang, Namyong Park, Thien Huu Nguyen, Jiebo Luo, Ryan A. Rossi, Julian McAuley
From Selection to Generation: A Survey of LLM-based Active Learning
Annual Meeting of the Association for Computational Linguistics (ACL), 2025.
40. Kaixiang Zhao, Lincan Li, Kaize Ding, Neil Zhenqiang Gong, [Yue Zhao](#), Yushun Dong
A Survey on Model Extraction Attacks and Defenses for Large Language Models
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD Lecture-Style Tutorial Track), 2025.
39. Shawn Li, Huixian Gong, Hao Dong, Tiankai Yang, Zhengzhong Tu, [Yue Zhao](#)
DPU: Dynamic Prototype Updating for Multimodal Out-of-Distribution Detection
Conference on Computer Vision and Pattern Recognition (CVPR),  **Highlight**, 2025
38. Hanhui Wang, Yihua Zhang, Ruizheng Bai, [Yue Zhao](#), Sijia Liu, Zhengzhong Tu
Edit Away and My Face Will Not Stay: Personal Biometric Defense against Malicious Generative Editing
Conference on Computer Vision and Pattern Recognition (CVPR), 2025
37. Yanbo Wang, Jiayi Ye, Siyuan Wu, Chujie Gao, Yue Huang, Xiuying Chen, [Yue Zhao](#), Xiangliang Zhang
TRUSTEVAL: A Dynamic Evaluation Toolkit on Trustworthiness of Generative Foundation Models
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL Demo Track), 2025.
36. Yuehan Qin[†], Yichi Zhang[†], Yi Nian[†], Xueying Ding, [Yue Zhao](#)
MetaOOD: Meta-learning for Automatic Out-of-Distribution Detection Model Selection
International Conference on Learning Representations (ICLR), 2025
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD Workshop on Resource-Efficient Learning for Knowledge Discovery),  **Best Paper Award**, 2024.
35. Sihan Chen, Zhuangzhuang Qian, Wingchun Siu, Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Tiankai Yang, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, [Yue Zhao](#)

- PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection
International World Wide Web Conference (The Web Conference Demo Track), 2025
34. Sizhe Liu, Yizhou Lu, Siyu Chen, Xiyang Hu, Jieyu Zhao, Tianfan Fu, Yue Zhao
DrugAgent: Automating AI-aided Drug Discovery Programming through LLM Multi-Agent Collaboration
AAAI Workshop on Foundation Models for Biological Discoveries (FMs4Bio), 2025.
 33. Hao Dong, Yue Zhao, Eleni Chatzi, Olga Fink
MultiOOD: Scaling Out-of-Distribution Detection for Multiple Modalities
Advances in Neural Information Processing Systems (NeurIPS),  **Spotlight**, 2024
 32. Xueying Ding, Yue Zhao, Leman Akoglu
Fast Unsupervised Deep Outlier Model Selection with Hypernetworks
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024
 31. Lichao Sun, Yue Huang, Haoran Wang, Siyuan Wu, Qihui Zhang, Chujie Gao, Yixin Huang, Wenhan Lyu, Yixuan Zhang, Xiner Li, Zhengliang Liu, Yixin Liu, Yijue Wang, Zhikun Zhang, 50+ collaborative authors, Yue Zhao
TrustLLM: Trustworthiness in Large Language Models
International Conference on Machine Learning (ICML), 2024
 30. Yue Zhao, Leman Akoglu
HPOD: Hyperparameter Optimization for Unsupervised Outlier Detection
International Conference on Automated Machine Learning (AutoML), 2024
 29. Yue Zhao
Towards Reproducible, Automated, and Scalable Anomaly Detection
AAAI Conference on Artificial Intelligence (AAAI), *New Faculty Highlights*, 2024
 28. Mingqi Jiang[†], Chaochuan Hou[†], Ao Zheng[†], Songqiao Han, Hailiang Huang[♣], Qingsong Wen, Xiyang Hu[♣], Yue Zhao[♣]
ADGym: Design Choices for Deep Anomaly Detection.
Advances in Neural Information Processing Systems (NeurIPS), 2023
([♣]Corresponding author)
 27. Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu
DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML/PKDD), 2023
 26. Peng Xu, Lin Zhang, Xuanzhou Liu, Jiaqi Sun, Yue Zhao, Haiqin Yang, Bei Yu
Do Not Train It: A Linear Neural Architecture Search of Graph Neural Networks
International Conference on Machine Learning (ICML), 2023
 25. Yue Zhao, Guoqing Zheng, Subhabrata Mukherjee, Robert McCann, Ahmed Awadallah
ADMoE: Anomaly Detection with Mixture-of-Experts from Noisy Labels
Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI), 2023
 24. Yue Zhao, George H. Chen, Zhihao Jia
TOD: GPU-accelerated Outlier Detection via Tensor Operations
International Conference on Very Large Data Bases (VLDB), 2023
 23. Songqiao Han[†], Xiyang Hu[†], Hailiang Huang[†], Mingqi Jiang[†], Yue Zhao^{†♣}
ADBench: Anomaly Detection Benchmark
Advances in Neural Information Processing Systems (NeurIPS), 2022
([†]Equal contribution & [♣]Corresponding author)
 22. Yue Zhao[†], Kay Liu[†], Yingdong Dou[†], et al.
Benchmarking Node Outlier Detection on Graphs
Advances in Neural Information Processing Systems (NeurIPS), 2022.
 21. Yue Zhao, Sean Zhang, Leman Akoglu
Toward Unsupervised Outlier Model Selection
IEEE International Conference on Data Mining (ICDM), 2022.
 20. Zhiming Xu, Xiao Huang, Yue Zhao, Yushun Dong, Jundong Li
Contrastive Attributed Network Anomaly Detection with Data Augmentation
Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2022.

19. [Yue Zhao](#), Ryan A. Rossi, Leman Akoglu
Automatic Unsupervised Outlier Model Selection
Advances in Neural Information Processing Systems (NeurIPS), 2021.
18. Kwei-Herng Lai, Daochen Zha, Junjie Xu, [Yue Zhao](#), Guanchu Wang, Xia Hu
Revisiting Time Series Outlier Detection: Definitions and Benchmarks
Advances in Neural Information Processing Systems (NeurIPS), 2021
17. Kexin Huang[†], Tianfan Fu[†], Wenhao Gao[†], [Yue Zhao](#), Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development
Advances in Neural Information Processing Systems (NeurIPS), 2021
16. [Yue Zhao](#)[†], Xiyang Hu[†], Cheng Cheng, Cong Wang, Changlin Wan, Wen Wang, Jianing Yang, Haoping Bai, Zheng Li, Cao Xiao, Yunlong Wang, Zhi Qiao, Jimeng Sun, Leman Akoglu
SUOD: Accelerating Large-scale Unsupervised Heterogeneous Outlier Detection
Conference on Machine and Learning Systems (MLSys), 2021.
15. Kwei-Herng Lai[†], Daochen Zha[†], Guanchu Wang, Junjie Xu, [Yue Zhao](#), Devesh Kumar, Yile Chen, Purav Zumkhawaka, Minyang Wan, Diego Martinez, Xia Hu
TODS: An Automated Time Series Outlier Detection System (Demo paper)
Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI), 2021.
14. Meng-Chieh Lee, [Yue Zhao](#), Aluna Wang, Pierre Jinghong Liang, Leman Akoglu, Vincent S. Tseng, Christos Faloutsos
AutoAudit: Mining Accounting and Time-Evolving Graphs
IEEE International Conference on Big Data (Big Data), 2020
13. Changlin Wan, Dongya Jia, [Yue Zhao](#), Wennan Chang, Sha Cao, Xiao Wang, and Chi Zhang
A Data Denoising Approach to Optimize Functional Clustering of Single Cell RNA-sequencing Data
IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2020
12. [Yue Zhao](#), Xueying Ding, Jianing Yang, Haoping Bai.
SUOD: Toward Scalable Unsupervised Outlier Detection
Workshops at the Thirty-Fourth AAAI Conference on Artificial Intelligence, 2020.
Extended version published in MLSys 2021.
11. Zheng Li, [Yue Zhao](#), Nicola Botta, Cezar Ionescu, Xiyang Hu
COPOD: Copula-Based Outlier Detection
IEEE International Conference on Data Mining (ICDM), 2020.
10. Zheng Li, [Yue Zhao](#), Jialin Fu
SYNC: A Copula based Framework for Generating Synthetic Data from Aggregated Sources
IEEE International Conference on Data Mining Workshops (ICDMW), 2020.
9. Yiqun Mei, [Yue Zhao](#), Wei Liang
DSR: An Accurate Single Image Super Resolution Approach for Various Degradations
IEEE International Conference on Multimedia and Expo (ICME), 2020, London, UK.
8. [Yue Zhao](#), Xuejian Wang[†], Cheng Cheng[†], Xueying Ding[†]
Combining Machine Learning Models and Scores using combo Library (Demo paper)
Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
7. Colin Wan, Zheng Li, Alicia Guo, [Yue Zhao](#)
SynC: A Unified Framework for Generating Synthetic Population with Gaussian Copula
Workshops at the Thirty-Fourth AAAI Conference on Artificial Intelligence, 2020.
Extended version published in ICDMW 2020.
6. Zain Nasrullah, [Yue Zhao](#)
Music Artist Classification with Convolutional Recurrent Neural Networks
IEEE International Joint Conference on Neural Networks (IJCNN), 2019, Hungary.
5. [Yue Zhao](#), Zain Nasrullah, Maciej K. Hryniewicki, Zheng Li
LSCP: Locally Selective Combination in Parallel Outlier Ensembles
SIAM International Conference on Data Mining (SDM), 2019, Calgary, Canada.
4. [Yue Zhao](#), Maciej K. Hryniewicki
DCSO: Dynamic Combination of Detector Scores for Outlier Ensembles
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD Workshop)

on *Outlier Detection De-constructed*), 2018, London, UK.

Extended version published in *SDM 2019*, renamed to *LSCP*.

3. [Yue Zhao](#), Maciej K. Hryniewicki
XGBOD: Improving Supervised Outlier Detection with Unsupervised Representation Learning
IEEE International Joint Conference on Neural Networks (IJCNN), 2018, Rio, Brazil.
2. [Yue Zhao](#), Maciej K. Hryniewicki, Francesca Cheng, Boyang Fu, Xiaoyu Zhu
Employee Turnover Prediction with Machine Learning: A Reliable Approach
Intelligent System Conference (Intellisys), 2018, London, UK.
1. [Yue Zhao](#)[†], Zhongtian Qiu[†], Yiqing Yang[†], Weiwei Li[†], Mingming Fan
An Empirical Study of Touch-based Authentication Methods on Smartwatches
ACM International Symposium on Wearable Computers (ISWC), 2017, Maui, USA.

INTERNSHIP EXPERIENCE	NortonLifeLock Research Group	
	Machine Learning Research Intern	2022
	Microsoft Research	
	Machine Learning Research Intern	2022
	Stanford University, Computer Science Department	
	Visiting Student Researcher (Prof. Jure Leskovec)	2021
	IQVIA, Analytics Center of Excellence	
	Machine Learning Research Intern	2020
	Siemens PLM Software USA	
	Software Engineer (Intern & Contract)	Mar. 2012 - Dec. 2014
TEACHING EXPERIENCE	University of Southern California	Los Angeles, CA
	Instructor	Spring 2027 (scheduled)
	<i>CSCI 699 Auditing AI Systems: Foundation Models and Agents</i>	
	Instructor	Fall 2026 (scheduled)
	<i>CSCI 566 Deep Learning and Its Applications</i>	
	Instructor	Spring 2026
	<i>CSCI 699 Adversarial and Trustworthy Foundation Models</i>	
	Instructor	Spring 2025
	<i>CSCI 566 Deep Learning and Its Applications</i>	
	Instructor	Spring 2024
	<i>CSCI 566 Deep Learning and Its Applications</i>	
	Carnegie Mellon University	Pittsburgh, PA
	Teaching Assistant	Fall 2022
	<i>Managing Digital Business</i> (Prof. David Riel)	
	Teaching Assistant & co-Instructor (lectures on AutoML and MLSys)	Spring 2022 – Fall 2020
	<i>Intro to Artificial Intelligence</i> (Prof. David Steier)	
	Teaching Assistant	Spring 2022
	<i>Digital Transformation</i> (Prof. David Riel)	
	Teaching Assistant (helping on course topics)	Fall 2021
	<i>Statistics for IT Managers</i> (Prof. Daniel Nagin)	
	University of Toronto	Toronto, ON
	Teaching Assistant & Lab Session Instructor	Fall 2015
	<i>Embedded Systems</i> (Prof. Philip Anderson)	
	University of Cincinnati	Cincinnati, OH
	Teaching Assistant & Lab Session Instructor	Fall 2014
	<i>Intro to Programming</i> (Prof. George Purdy)	

PH.D. STUDENTS

- Haoyan Xu (USC, ECE Ph.D. Candidate, 2024 Spring-), co-advised by Mengyuan Li,  Capital One Fellowship
- Yuehan Qin (USC, CS Ph.D. Candidate, 2024 Fall-)
- Tiankai Yang (USC, CS Ph.D., 2024 Fall-)
- Shawn Li (USC, CS Ph.D. Candidate, 2024 Fall-),  Capital One Fellowship, Amazon ML Fellowship
- Jiate Li (USC, CS Ph.D., 2025 Fall-)
- Wei Yang (USC, CS Ph.D. Candidate, 2026 Fall-), co-advised by Jesse Thomason and Xuezhe Ma
- Yi Nian (USC, CS Ph.D., incoming)
- Chenxiao Yu (USC, CS Ph.D., 2026 Fall-)
- Zixiang Xu (USC, CS Ph.D., 2026 Fall-)

STUDENT COMMITTEE	Term	Student	Committee
	Fall 2026	Chen Chu (USC, CS Ph.D.)	Qualification
		Yuqing Yang (USC, CS Ph.D.)	Qualification
		Zeyu Shanguan (USC, CS Ph.D.)	Qualification
		Chufan Shi (USC, CS Ph.D.)	Qualification
		Linxin Song (USC, CS Ph.D.)	Qualification
		Raja Kumar (USC, CS Ph.D.)	Qualification
		Ziyi Liu (USC, CS Ph.D.)	Thesis
	Summer 2026	Defu Cao (USC, CS Ph.D.)	Defense
	Spring 2026	Longchao Da (ASU, CS Ph.D.)	Defense
		Yahan Li (USC, CS Ph.D.)	Qualification
		Jiageng Mao (USC, CS Ph.D.)	Thesis
		Xinyue Cui (USC, CS Ph.D.)	Qualification
		Jiawei Yang (USC, CS Ph.D.)	Qualification + Thesis
		Elnaz Rahmati (USC, CS Ph.D.)	Qualification
	Fall 2025	Hao Dong (ETH Zurich, CS Ph.D.)	Defense
		Haonan Wang (USC, ECE Ph.D.)	Defense
	Spring 2025	Mengxi Wu (USC, CS Ph.D.)	Qualification
		Xingrui Wang (USC, ME Ph.D.)	Defense
	Fall 2024	Alex Bisberg (USC, CS Ph.D.)	Thesis
	Summer 2024	Gengyu Rao (USC, CS Ph.D.)	Defense
		Mehrdad Kiamari (USC, ECE Ph.D.)	Defense
	Spring 2024	Maria Despoina Siampou (USC, CS Ph.D.)	Qualification
		Yuan Meng (USC, ECE Ph.D.)	Defense
		Hassan Hamad (USC, ECE Ph.D.)	Qualification
		Yizhou Zhang (USC, CS Ph.D.)	Thesis
		Haoming Li (USC, CS Ph.D.)	Qualification
		Arash Hajisafi (USC, CS Ph.D.)	Qualification
	Fall 2023	Yi Chien Lin (USC, ECE Ph.D.)	Qualification
		Yuke Zhang (USC, ECE Ph.D.)	Qualification

SERVICES

Institutional Leadership

- Associate Co-Director, USC Institute on Ethics and Trust in Computing (IETC), 2026–2027

Community Outreach

- Founder & Maintainer, CSpHD.org: a non-profit community for CS/AI/EE/Stats PhDs (since 2022, several thousand members, 500+ PhD/RA/postdoc openings), supporting students from before enrollment to the faculty job market.

Conference/Workshop Organizing Roles

- Co-organizer, Geometric and Spatial Reasoning in Vision-Language Models (GSR-VLM) Workshop @ AAAI 2027
- Doctoral Consortium Co-Chair, AAAI 2027
- Future Faculty Highlights Co-Chair, IEEE BigData 2026
- Publicity Co-Chair, WSDM 2027
- Workflow Co-Chair, KDD 2023
- Co-organizer, SURGeLLM: Structured Understanding, Retrieval, and Generation in LLMs era Workshop @ ACL 2026
- Co-organizer, AgentSkills '26: The First Workshop on Agent Skills @ CAIS Conference 2026
- Co-organizer, AI for Financial Fraud Detection & Prevention Workshop @ 6th ACM International Conference on AI in Finance

Funding Proposal Review Roles

- National Science Foundation (NSF)
- Dutch Research Council (NWO)

Journal Editorial Roles

- Action Editor, Transactions on Machine Learning Research (TMLR), 2026–present
- Associate Editor, ACM Transactions on AI for Science (TAIS), 2025–present
- Associate Editor, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024–present
- Action Editor, Journal of Data-centric Machine Learning Research (DMLR), 2024–present

Conference/Workshop AC and Reviewer Roles

- ICLR 2025 (AC), ICLR 2026 (AC)
- AAAI 2021, 2022, 2023, 2025 (Senior PC), 2026 (Senior PC)
- ICML 2024, 2025 (AC), 2026 (AC)
- NeurIPS 2021, 2022, 2023, 2025 (AC), 2026 (AC)
- AISTATS 2024, 2025 (AC)
- MLSys 2024, 2026
- KDD 2020, 2021, 2022, 2023, 2026
- ACL Rolling Review (ARR) 2026
- IJCAI 2022, 2023
- AAAI Demonstrations 2021, 2022
- MICCAI 2020, 2021, 2022
- ICDM 2020
- KDD Workshop on Outlier Detection and Description (ODD), 2021
- KDD Workshop on Anomaly and Novelty Detection (ANDEA), 2021, 2022
- IJCAI Workshop on Artificial Intelligence for Anomalies and Novelty (AI4AN), 2020, 2021
- INFORMS Workshop on Data Science 2021

Journal Review Roles

- Journal of Machine Learning Research (JMLR)
- PNAS Nexus
- Machine Learning
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Intelligent Systems
- IEEE Journal on Selected Areas in Communications (J-SAC)
- Data Mining and Knowledge Discovery (DMAI)

- ACM Transactions on Management Information Systems (TMIS)
- Knowledge and Information Systems (KAIS)
- INFORMS Journal on Computing (IJOC)
- Big Data
- Artificial Intelligence Review (AIRE)
- Neurocomputing
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)
- IEEE Network Magazine
- IEEE Computational Intelligence Magazine (CIM)
- BioData Mining
- European Journal of Management and Business Economics (EJMBE)
- The Journal of Open Source Software (JOSS)

TALKS AND LECTURES	USC IETC Reading Group	<i>Safe Models, Unsafe Agents: From Defense Training to Agent Auditing</i>	Sep. 2026
	Arizona State University	Guest Lecture <i>My Unexpected Programming Journey</i>	Sep. 2026
	NVIDIA Agent & Security	<i>Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker</i>	May. 2026
	Amazon AI Security	<i>Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker</i>	May. 2026
	USC symposium on Frontiers of ML/AI	<i>Towards Robust AI: Advances in Outlier and OOD Detection</i>	Mar. 2025
	NUS Tea Talk	<i>Towards Robust AI: Advances in Outlier and OOD Detection</i>	Jan. 2025
	SFU@NeurIPS'24	<i>Towards Robust AI: Advances in Outlier and OOD Detection</i>	Dec. 2024
	KAIST	<i>Unsupervised Model Selection: Automation with Meta-learning and LLMs</i>	Nov. 2024
	Kennesaw State University	<i>Unsupervised Model Selection: Automation with Meta-learning and LLMs</i>	Oct. 2024
	LinkedIn Anti-Abuse AI	<i>Outlier Detection: Automation, Systems, and GenAI</i>	Aug. 2024
	Amazon Security AI	<i>Outlier Detection: Automation, Systems, and GenAI</i>	Aug. 2024
	New York University	<i>Outlier Detection: Automation, Systems, and GenAI</i>	Aug. 2024
	University of Washington	<i>Outlier Detection: Automation, Systems, and GenAI</i>	Jun. 2024
	Microsoft	<i>Outlier Detection: Automation, Systems, and GenAI</i>	Jun. 2024
	USC Retreat on AI and Engineering Safety	<i>Safety Measures for LLMs</i>	Apr. 2024
	Visa Research	<i>Towards Reproducible, Automated, and Scalable AD</i>	Apr. 2024
	USC Symposium on Frontiers of Generative AI	<i>Generative AI for Anomaly Detection</i>	Mar. 2024
	AAAI New Faculty Highlights (invited)	<i>Towards Reproducible, Automated, and Scalable AD</i>	Feb. 2024
	U of Nevada, Las Vegas	<i>Automated and Scalable ML Algorithms and Systems</i>	Oct. 2023
	Samsung Seminar	<i>Automated and Scalable Anomaly Detection Systems</i>	Aug. 2023
	KDD SoCal Day	<i>Enable Applications by ML with Noisy Inputs</i>	Aug. 2023
	CMU Catalyst	<i>How (Not) to Fail Your Academic Job Search</i>	May. 2023
	KAUST	<i>Automated and Scalable ML Algorithms and Systems</i>	Apr. 2023
	Emory University	<i>Automated and Scalable ML Algorithms and Systems</i>	Apr. 2023
	USC	<i>Automated and Scalable ML Algorithms and Systems</i>	Mar. 2023
	UC Davis	<i>Automated and Scalable ML Algorithms and Systems</i>	Mar. 2023
	Stony Brook University	<i>Automated and Scalable ML Algorithms and Systems</i>	Feb. 2023
	University of Chicago	<i>Automated and Scalable ML Algorithms and Systems</i>	Feb. 2023
	UC Merced	<i>Automated and Scalable ML Algorithms and Systems</i>	Feb. 2023
	CMU PDL Meeting	<i>Automated and Scalable ML Algorithms and Systems</i>	Jan. 2023
	CMU Data Science Seminar	Guest Lecture <i>Automated Anomaly Detection</i>	Nov. 2022
	LoG Seminar	<i>Large-scale Graph Anomaly Detection</i>	Oct. 2022
	Intuit	<i>Anomaly Detection for Financial Risk Modeling</i>	Aug. 2022
	Rice University	<i>Large-scale Anomaly Detection with Automation</i>	Sep. 2022
	Microsoft Research	<i>Weakly-supervised Anomaly Detection</i>	Sep. 2022
	Wells Fargo	<i>Anomaly Detection for Financial Risk Modeling</i>	Aug. 2022
	Columbia University	Guest Lecture <i>Anomaly Detection</i>	Jul. 2022
	Morgan Stanley	<i>Automated Outlier Detection</i>	Jun. 2022
	Microsoft Research	<i>Automated Outlier Detection</i>	Jun. 2022
	Morgan Stanley	<i>Large-scale Anomaly Detection Systems</i>	Mar. 2022
	Rutgers Business School	<i>Outlier Model Selection</i>	Mar. 2022
	Tesla	<i>Large-scale Anomaly Detection Systems</i>	Feb. 2022
	Catalyst, CMU	<i>Systems for Data Mining Algorithms</i>	Dec. 2021
	E&Y Canada	<i>ML applications in Data Analytics</i>	Oct. 2021
	University of Nottingham	<i>General Machine Learning Applications</i>	Jan. 2021